Evaluating a multi-dimensional noise model for differential expression in multi-tissue RNA-seq data

Student project kick-off

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Agenda

1. Scientific question
2. Material: biology and the data we use
3. Methods: the noise model and how to evaluate it
4. Data: what is available, what is expected
5. First steps: the first two weeks
6. Outlook: the coming months

Scientific question

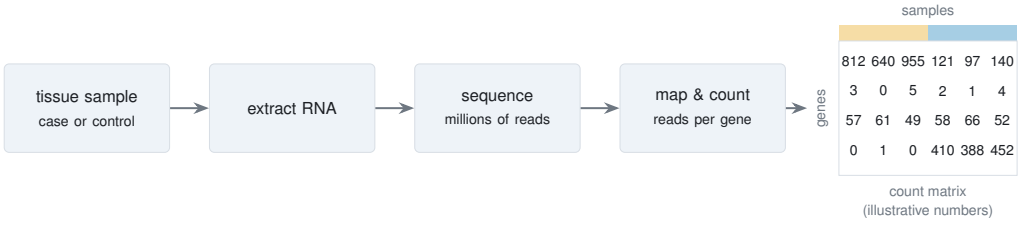
Does the multi-dimensional TOX model control false positives,
and how does its power compare with limma, edgeR and DESeq2?

Calibration: are the p-values honest?

Power: does it find true effects?

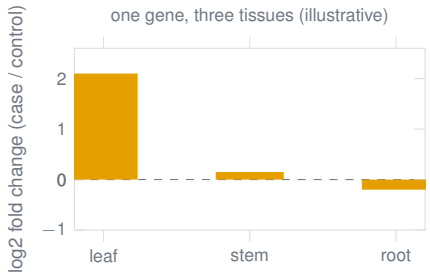
Biology primer: gene expression and RNA-seq

- gene $\rightarrow$ RNA $\rightarrow$ protein; expression = how much RNA
- RNA-seq: one read **count** per gene per sample
- replicates: independent samples of the same condition
- sequencing depth differs between samples $\Rightarrow$ normalization



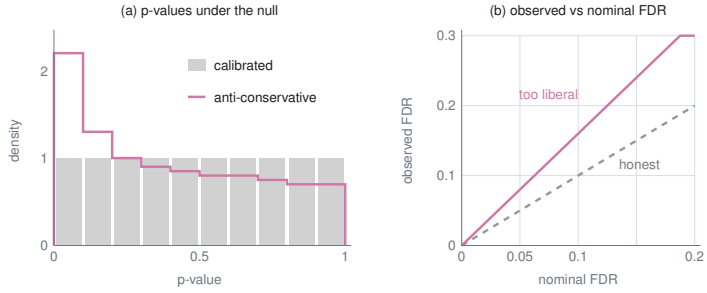
Case vs control, across tissues

- contrast: **case** vs **control**
- effect: log2 fold change (log2FC)
- one gene, d tissues $\Rightarrow$ a **vector** of effects
- question: does the gene respond *anywhere*?



Statistics primer

- null distribution
- p-value
- calibration
- FDR (Benjamini–Hochberg)
- power (TPR)



illustrative shapes, not data

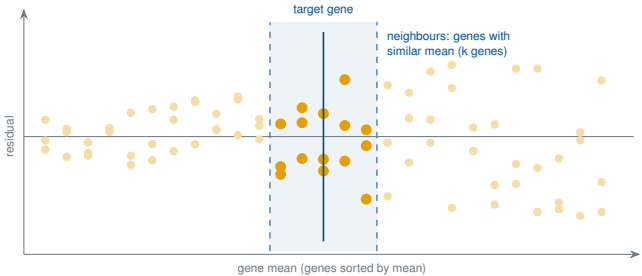
Existing tools

Tool	Noise model	Variance estimate
edgeR	negative binomial on counts	dispersion, empirical Bayes shrinkage
DESeq2	negative binomial on counts	dispersion, shrunk towards a mean trend
limma	linear model on log values	empirical Bayes moderation

- multi-tissue: **no obvious 1:1 reference** – part of your task

The TOX idea (one tissue)

- estimate the noise **from the data**
- neighbours: genes with a similar mean
- pool their residuals
- read the null off the pool



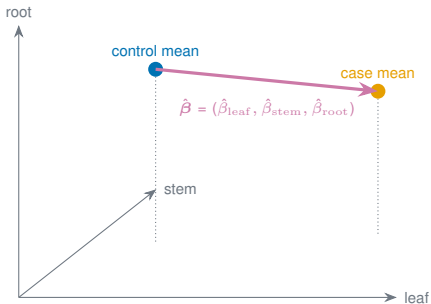
schematic: one group (e.g. case); not data

1-D mechanics

1. group means per gene (case, control)
2. residual pool: adaptive kNN, counted in **genes**
3. scale each pool by $1/\sqrt{n}$ of its own group
4. null of the mean difference: all case–control residual pairs
5. p-value from the tail count

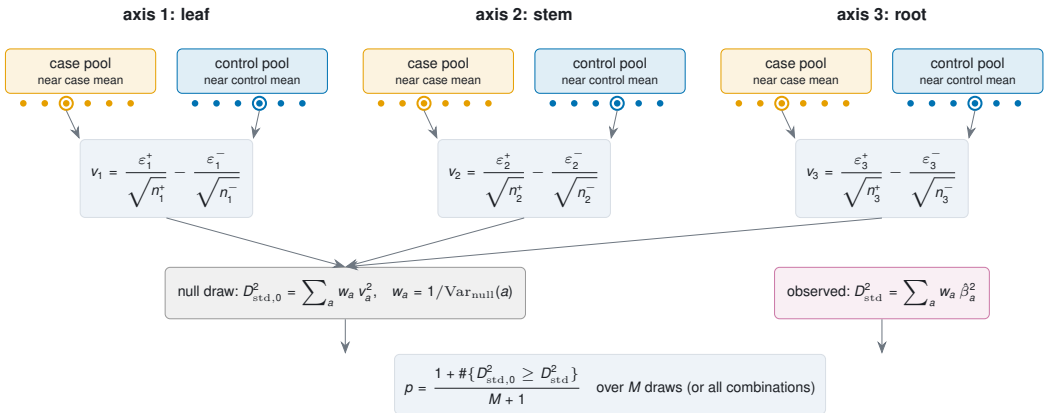
The multi-tissue problem

- gene = one vector over *d* tissues (*axes*)
- axes treated as **independent**
- no sample shared between axes
- R wrapper checks sample IDs (`check_axis_samples`)



The multi-dimensional model

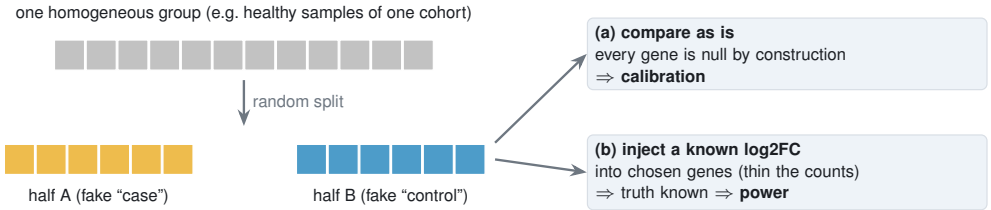
$\hat{\beta}_a = \bar{x}_a^{\text{case}} - \bar{x}_a^{\text{ctrl}}$, $D = \sqrt{\sum_a \hat{\beta}_a^2}$ (effect size), $D_{\text{std}} = \sqrt{\sum_a \hat{\beta}_a^2 / \text{Var}_{\text{null}}(a)}$ (test statistic)



Assumptions, open questions, what to look for

1. genes with similar group means have exchangeable within-group residuals
2. axes share no samples
3. **open:** a noise scale shared across tissues is not captured
4. **open:** sensitivity to k, τ, normalization

Evaluation design

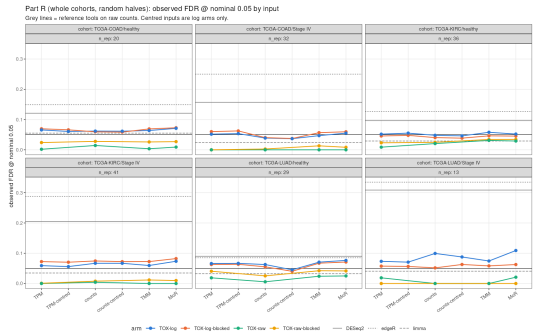


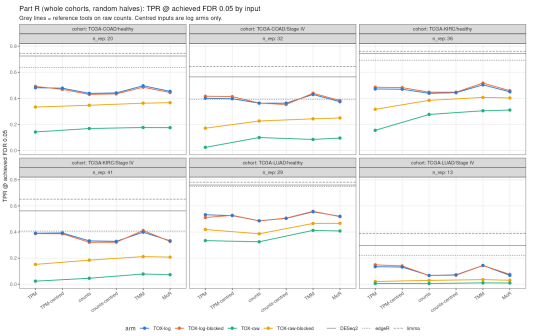
Metrics and plots

- QQ plot / histogram of null p-values
- false-positive rate at 0.001, 0.01, 0.05
- observed vs nominal FDR
- TPR at observed (achieved) FDR

Where the 1-D model stands today

scalar model, TCGA random halves with injected effects (Part R), 20 rounds





[TODO: presenter: confirm before showing; run of 19 Oct 2026]

Material and data

- **available:** TCGA RNA-seq; healthy samples, one cohort per axis (e.g. TCGA-KIRC, TCGA-LUAD, TCGA-STAD)
- **available:** existing scripts for calibration and power
- **highly recommended:** a self-assembled multi-tissue dataset
- **not as main data:** mammal tissue set in `material/`

Criteria for a new dataset

- enough replicates per tissue to split (roughly ≥ 6)
- independent samples across tissues
- homogeneous samples within a tissue
- count data available
- combined studies: same platform and quantification

What you do, in tracks

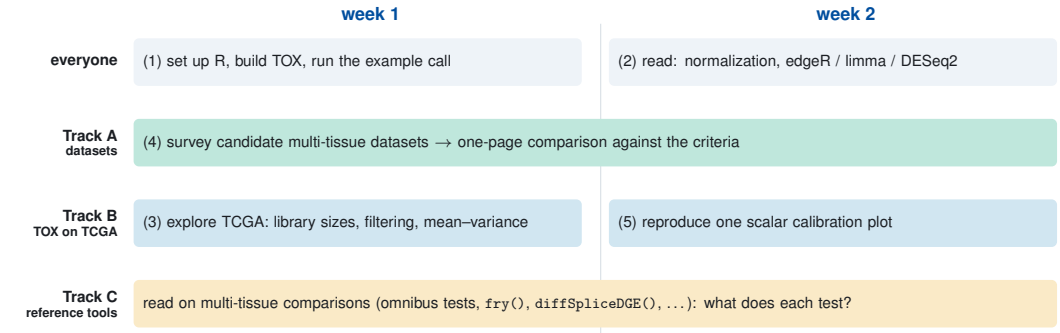
- **Track A:** dataset survey and assembly
- **Track B:** TOX on TCGA, calibration scripts
- **Track C:** reference tools, metrics, multi-tissue comparison
- **shared:** onboarding, final report

Repository tour

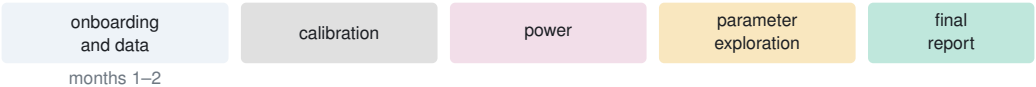
overview	`README.md`, `concise_project_info.md`
manuscript	`misc/Tensor_Omics_Methods.pdf`
R interface	`rcpp/tensoromics_functions.R`
tests	`test/mod_test_noise_model*.f90`
docs	`experiments/Noise_Model_Test/docs/`
test scripts	`experiments/Noise_Model_Test/`

```
res <- tox_compute_noise_pvalues_pipeline_md(
  case_replicates    = list(leaf = case_leaf,
                            root = case_root),
  control_replicates = list(leaf = ctrl_leaf,
                            root = ctrl_root),
  max_pool_size      = 50000L)
```

First two weeks



Outlook



- calibration first, power as time allows
- about 180 h per student (6 ECTS): a guideline, not a target

Questions?

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TAKEAWAYS

- the noise model decides every DE result
- TOX: empirical null; check where its assumptions fail
- your work: calibration and power, in parallel tracks

Backup

B1: Null construction, worked example for $d = 2$

- pool per axis and side: k genes $\times$ n replicates residuals
- e.g. $k = 20$, $n = 3$: 60 per side, $S_a = 60 \times 60 = 3600$ pairs
- joint space $S_1 S_2 \approx 1.3 \times 10^7 \Rightarrow$ sampled
- weights $w_a = 1/\mathrm{Var}_{\mathrm{null}}(a)$
- check: $\mathrm{E}[D^2_{\mathrm{std},0}]$ = number of axes with $w_a > 0$

B2: Three p-value paths and resolution

- enumeration: all combinations, while $\prod_a S_a \leq \text{cap}$
- systematic stride sampling (default): deterministic, no seed
- random sampling: classical Monte Carlo
- floor $1/(M+1)$; current default $M = 20{,}000$
- M set by the gene count under BH, not by d

B3: Parameters (current defaults, may change)

Argument	Current default	Meaning
`k_start`, `k_step`, `k_max`	20, 1, 50	neighbour genes
`tau`	0.1	stop growing the pool
`norm_method`	1 (log2)	0 = linear
`beta_centre`	0	1 = subtract per-axis median $\hat{\beta}$
`n_draws_max`, `sampling_mode`	20000, 0	M; systematic

B4: Why scale residuals by $1/\sqrt{n}$

- observed: a difference of **means** over n replicates
- one residual: single-observation scale
- $\mathrm{Var}(\bar{x}) = \sigma^2/n \Rightarrow$ scale by $1/\sqrt{n}$
- without it: null too wide, test conservative
- caveat: fixes the variance, not the *shape*

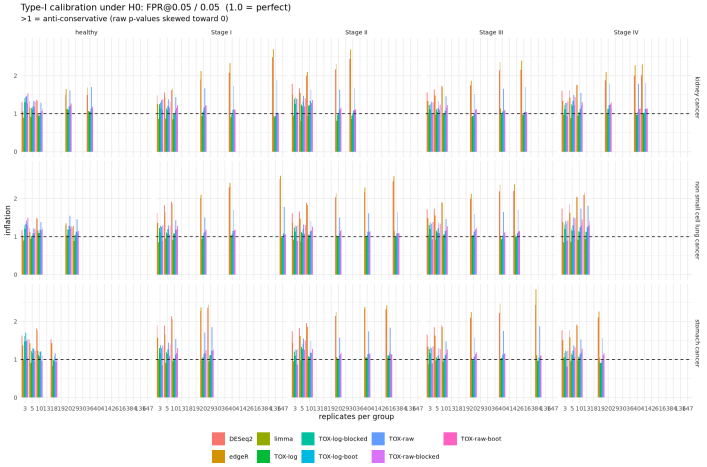
B5: Why not a joint neighbourhood in *d* dimensions?

- joint kNN: neighbours must match on *all* axes at once
- distances grow with *d*: fewer good matches
- tissue-specific genes match poorly on at least one axis
- needed only if axes share samples (out of scope)

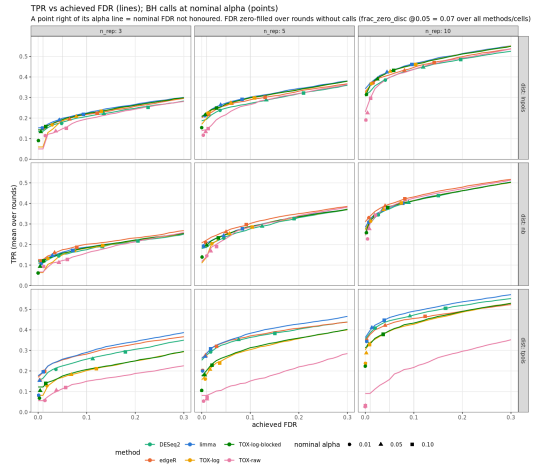
B6: Glossary

axis	one tissue (or cohort) dimension of a gene's effect vector
$\hat{\beta}_a$	case minus control mean on axis a
D, D_{std}	effect size; standardized test statistic
residual	replicate minus its group mean
neighbourhood	genes with similar group mean, per axis and side
calibration	FPR at a threshold equals the threshold
FDR, TPR	false discoveries among calls; share of true effects found

B7: Scalar model, mock-null calibration on TCGA



B8: Scalar model, simulated counts (Part P)



B9: Scalar model, Part R precision–recall

